Unit 8: Ecology

What's the big deal?

Interactions between complex living organisms can lead to changes in communities and ecosystems.

Pages 1-5 are meant to help highlight the important information included in each unit. While this can be a major source of information, make sure to use any/all studying tools available to you (ie class notes, textbook, videos, supplemental study books, etc.) Please note that while the College Board lays out the topics in this order, the class may go through the topics in a different order or combine similar topics.

Topic 1: Responses to the Environment

Learning Objective

ENE-3.D: Explain how the behavioral and/or physiological response of an organism is related to changes in internal or external environment.

IST-5.A: Explain how the behavioral responses of organisms affect their overall fitness and may contribute to the success of the population.

Essential Knowledge

ENE-3.D.1: Organisms respond to changes in their environment through behavioral and physiological mechanisms.

ENE-3.D.2: Organisms exchange information with one another in response to internal changes and external cues, which can change behavior.

IST-5.A.1: Individuals can act on information and communicate it to others.

IST-5.A.2: Communication occurs through various mechanisms—

a. Organisms have a variety of signaling behaviors that produce changes in the behavior of other organisms and can result in differential reproductive success.
b. Animals use visual, audible, tactile, electrical, and chemical signals to indicate dominance, find food, establish territory, and ensure reproductive success.

IST-5.A.3: Responses to information and communication of information are vital to natural selection and evolution—

a. Natural selection favors innate and learned behaviors that increase survival and reproductive fitness.
b. Cooperative behavior tends to increase the fitness of the individual and the survival of the population.

Vocabulary/Review Questions

Ethology	Behavior	Proximate cause
Innate behaviors	Learned behaviors	Ultimate cause
Fixed Action Patterns	Stimulus response chain	Directed movements
Pheromones	Sign stimulus	Migration
Kinesis	Taxis	Phototaxis
Chemotaxis	Geotaxis	Learning
Imprinting	Spatial learning	Cognitive map
Associative learning	Social learning	Foraging
Altruism	Phototropism	Photoperiodism

1. What is the difference between an innate behavior and a learned behavior? Research an example of each (not from the notes) and discuss each one.
2. When being pet, cats tend to purr. What kind of learning does this represent?
3. What does the concept of "nature vs. nurture" describe?
4. How do stickleback fish exhibit fixed action patterns?
5. How do bees communicate to other bees in their hive the location of food?
6. Learned behaviors always superior to innate behaviors. Do you support this statement? Why or why not?
7. Why do some organisms show altruistic behavior?
8. Describe the ways in which organisms communicate with each other.

Topic 2: Energy Flow Through Ecosystems

Learning Objective

ENE-1.M: Describe the strategies organisms use to acquire and use energy.

ENE-1.N: Explain how changes in energy availability affect populations and ecosystems.

ENE-1.O: Explain how the activities of autotrophs and heterotrophs enable the flow of energy within an ecosystem

Essential Knowledge

ENE-1.M.1: Organisms use energy to maintain organization, grow, and reproduce—

Vocabulary/Review Questions

Metabolic rate	Ecosystem	Biotic
Abiotic	Endotherm	Ectotherm
Primary producer	Heterotrophs	Primary consumer
Secondary consumer	Tertiary consumer	Decomposer
Food chain	Food web	Primary production
GPP	NPP	Secondary production

a. Organisms use different strategies to regulate body temperature and metabolism
   i. Endotherms use thermal energy generated by metabolism to maintain homeostatic body temperatures.
   ii. Ectotherms lack efficient internal mechanisms for maintaining body temperature, though they may regulate their temperature behaviorally by moving into the sun or shade or by aggregating with other individuals.

b. Different organisms use various reproductive strategies in response to energy availability.

c. There is a relationship between metabolic rate per unit body mass and the size of multicellular organisms—generally, the smaller the organism, the higher the metabolic rate.

d. A net gain in energy results in energy storage or the growth of an organism.

e. A net loss of energy results in loss of mass and, ultimately, the death of an organism.

ENE-1.N.1: Changes in energy availability can result in changes in population size.

ENE-1.N.2: Changes in energy availability can result in disruptions to an ecosystem—

a. A change in energy resources such as sunlight can affect the number and size of the trophic levels.

b. A change in the producer level can affect the number and size of other trophic levels.

ENE-1.O.1: Autotrophs capture energy from physical or chemical sources in the environment—

a. Photosynthetic organisms capture energy present in sunlight.

b. Chemosynthetic organisms capture energy from small inorganic molecules present in their environment, and this process can occur in the absence of oxygen.

ENE-1.O.2: Heterotrophs capture energy present in carbon compounds produced by other organisms.

a. Heterotrophs may metabolize carbohydrates, lipids, and proteins as sources of energy by hydrolysis.

1. How does the number of links in a food web correlate to the overall stability of the community?
2. Link the laws of thermodynamics to a discussion on the flow of energy through the trophic levels (producer to tertiary consumer).
3. Discuss the role of photosynthesis and cellular respiration in carbon cycling in the biosphere.
4. Think back to Unit 1, what makes carbon such a versatile element?
5. Discuss how energy flows through an ecosystem and the relative efficiency with which it occurs.
6. What do arrows represent in a food chain/web?
7. True or false: smaller organisms have a higher metabolic rate.
8. How would an event, like deforestation, affect the water cycle?
9. How does climate change affect the carbon cycle?
10. Fill in the blank: The transfer of energy between trophic levels is at around ______ efficiency.
11. Why are there not an infinite number of trophic levels? Have you noticed that there are only ever about 5? Why is this?
12. Fill in the blank: the bottom of most trophic levels is represented by ___________.
13. Think back to Unit 3. Describe the process of photosynthesis.
14. In reference to the previous question, how do plants convert the sun's energy into a form that can be used by heterotrophs?
15. What would happen to the stability of a community if an organism is removed?

Topic 3: Population Ecology

Learning Objective

SYI-1.G: Describe factors that influence growth dynamics of populations.

Essential Knowledge

SYI-1.G.1: Populations comprise individual organisms that interact with one another and with the environment in complex ways.

SYI-1.G.2: Many adaptations in organisms are related to obtaining and using energy and matter in a particular environment—

a. Population growth dynamics depend on a number of factors.
   i. Reproduction without constraints results in the exponential growth of a population.

Vocabulary/Review Questions

Demography	Life table	Survivorship curve
Exponential growth	Life history	Density-dependent regulation
Density-independent regulation	K-selection	R-selection

1. List three factors that can affect how quickly a population will grow.
2. Describe the three types of survivorship curves.
3. In reference to the previous question, research an organism that represents each type of survivorship curve.

Relevant Equation:
Population Growth-

$$dN/dt = B - D$$

where:
dt = change in time
B = birth rate
D = death rate
N = population size

Relevant Equation:
Exponential Growth-

$$dN/dt = r_{max}N$$

where:
dt = change in time
N = population size
r_{max} = maximum per capita growth rate of population

4. What type of survivorship curve would represent humans living in the United States?
5. Circle the correct options: Imagine a hypothetical population that has all of its needs met and unlimited resources. In this population the rate of births would be low/high and the rate of deaths would be low/high.
6. Would global climate change have more of an impact on density dependent factors or density independent factors? Why?
7. Would a disease that spreads through a population have more of an impact on density dependent factors or density independent factors? Why?

Topic 4: Effect of Density of Populations

Learning Objective

SYI-1.H: Explain how the density of a population affects and is determined by resource availability in the environment.

Essential Knowledge

SYI-1.H.1: A population can produce a density of individuals that exceeds the system's resource availability.
SYI-1.H.2: As limits to growth due to density-dependent and density-independent factors are imposed, a logistic growth model generally ensues.

Relevant Equation:

$$dN/dt = r_{max}N[(K-N)/K]$$

where:
dt = change in time
N = population size
r_{max} = maximum per capita growth rate of population
K = carrying capacity

Vocabulary/Review Questions

Population	Population ecology	Density
Dispersion	Logistic growth	

1. What are the three dispersion patterns? Draw an illustration of each.
2. List two reasons why clumped dispersion is commonly seen in nature.
3. What is the difference between the size of the population and its density?
4. In terms of the human population, how can ecologists use demographic analysis and the prediction of future population growth to help governments/ schools/ hospitals/ etc prepare for needs?
5. Research one species that represents an example of logistic population growth. Describe the species
6. What happens to resources as a population grows in a logistic growth model?

Topic 5: Community Ecology

Learning Objective

ENE-4.A: Describe the structure of a community according to its species composition and diversity
ENE-4.B: Explain how interactions within and among populations influence community structure.
ENE-4.C: Explain how community structure is related to energy availability in the environment.

Essential Knowledge

ENE-4.A.1: The structure of a community is measured and described in terms of species composition and species diversity.

Relevant Equation:
Simpson's Diversity Index-

$$\text{Diversity Index} = 1 - \sum (n/N)^2$$

where:
n = the total number of organisms of a particular species
N = total number of organisms of all species

Vocabulary/Review Questions

Community	Habitat	Ecological niche
Fundamental niche	Realized niche	Interspecific interactions
Competition	Niche partitioning	Predation
Herbivory	Symbiosis	Parasitism
Mutualism	Commensalism	Facilitation
Biodiversity	Species richness	Relative abundance

1. What is the difference between a population, a community, an ecosystem, a biome, and a biosphere?
2. How does an ecological niche differ from a fundamental niche?

ENE-4.B.1: Communities change over time depending on interactions between populations.
ENE-4.B.2: Interactions among populations determine how they access energy and matter within a community
ENE-4.B.3: Relationships among interacting populations can be characterized by positive and negative effects and can be modeled. Examples include predator/prey interactions, trophic cascades, and niche partitioning.
ENE-4.B.4: Competition, predation, and symbioses, including parasitism, mutualism, and commensalism, can drive population dynamics.
ENE-4.C.1: Cooperation or coordination between organisms, populations, and species can result in enhanced movement of, or access to, matter and energy.

3. What does the competitive exclusion principle state?
4. In general, describe the physiological and behavioral adaptations of predators.
5. On your own, find one organism of interest to you. Research the adaptations this organism has to avoid predation. Record your findings here.
6. Compare and contrast the different type of symbiosis.
7. Describe the relationship that humans have with 5 other organisms (choose organisms we do not eat).
8. True or false: a +/- relationship describes predation.
9. How does Batesian mimicry differ from Mullerian mimicry?

Topic 6: Biodiversity

Learning Objective

SYI-3.F: Describe the relationship between ecosystem diversity and its resilience to changes in the environment.
SYI-3.G: Explain how the addition or removal of any component of an ecosystem will affect its overall short-term and long-term structure.

Essential Knowledge

SYI-3.F.1: Natural and artificial ecosystems with fewer component parts and with little diversity among the parts are often less resilient to changes in the environment.
SYI-3.F.2: Keystone species, producers, and essential abiotic and biotic factors contribute to maintaining the diversity of an ecosystem
SYI-3.G.1: The diversity of species within an ecosystem may influence the organization of the ecosystem.
SYI-3.G.2: The effects of keystone species on the ecosystem are disproportionate relative to their abundance in the ecosystem, and when they are removed from the ecosystem, the ecosystem often collapses.

Vocabulary/Review Questions

Biodiversity	Keystone species

1. Research one example of a terrestrial keystone species. Why is it considered a keystone species? Record your research here.
2. What would happen to a community if a keystone species were to be removed?
3. Keystone species make the community stronger and more resilient to change. Do you agree with this statement? Why or why not?
4. List three reasons for why biodiversity should be maintained in the biosphere.
5. How can maintaining biodiversity be helpful to humans?
6. Imagine you are a conservation biologist studying monk seals. What information/data would you gather about these animals in order to create a conservation plan?

Topic 7: Disruptions to Ecosystems

Learning Objective

EVO-1.O: Explain the interaction between the environment and random or preexisting variations in populations.
SYI-2.A: Explain how invasive species affect ecosystem dynamics.
SYI-2.B: Describe human activities that lead to changes in ecosystem structure and/ or dynamics.
SYI-2.C: Explain how geological and meteorological activity leads to changes in ecosystem structure and/or dynamics.

Essential Knowledge

EVO-1.O.1: An adaptation is a genetic variation that is favored by selection and is manifested as a trait that provides an advantage to an organism in a particular environment.
EVO-1.O.2: Mutations are random and are not directed by specific environmental pressures.

Vocabulary/Review Questions

Ecological succession	Primary succession	Secondary succession
Disturbance	Habitat loss	Overharvesting
Global change	Invasive species	Biogeographical factors
Pathogens		

1. Compare and contrast primary and secondary succession.
2. How do the extinction rates of today compare to the extinction rates of the past? You may use online resources to help you with this question.
3. Think back to Unit 7. What is a bottleneck effect? How can a bottleneck effect increase the chance of a species becoming extinct?

SYI-2.A.1: The intentional or unintentional introduction of an invasive species can allow the species to exploit a new niche free of predators or competitors or to outcompete other organisms for resources. SYI-2.A.2: The availability of resources can result in uncontrolled population growth and ecological changes. SYI-2.B.1: The distribution of local and global ecosystems changes over time. SYI-2.B.2: Human impact accelerates change at local and global levels— a. The introduction of new diseases can devastate native species. b. Habitat change can occur because of human activity SYI-2.C.1: Geological and meteorological events affect habitat change and ecosystem distribution. Biogeographical studies illustrate these changes.	4. Cloning technology has advanced in recent years. Do you believe cloning endangered animals would resolve their issue of endangerment? Why or why not? 5. Research carbon dioxide. How do increased atmospheric levels affect oceans? How would this affect life in the ocean? 6. To meet the growing demands of the population, many crops are being genetically modified. What does this mean and what is one benefit and one risk of using genetically modified organisms? *You may use outside sources to help answer this question.* 7. Discuss at least two factors that lead to the success of an invasive species in its new habitat. 8. List three reasons for why invasive species are dangerous to the environment.

Topic 1

Behaviors

Think, pair, share:

1. How do behaviors in animals arise?

2. What purpose do they serve?

Behavioral Ecology

In the 1970s a group of scientists won the Nobel Prize in Physiology for their discoveries concerning organization and elicitation of ________________________ and ________________ behavior patterns

- These studies developed the discipline of ethology

Ethology:

Behavior:

- _______________ vs ______________ (genetic and environmental factors)
- Allow for ___________________ and ________________________
  - Subject to:

➔ Questions?
➔ Textbook chapters/pages to review

Understanding Behavior

Proximate cause:

1. What was the ________________ to __________ the behavior?

2.

(how do the experiences during growth and development influence the response)

Ultimate cause:

1. How does the behavior help the animal _________________ and _____________________?

2.

(what is the evolutionary basis of the behavior)

Practice Problem

1. While drinking water may not seem like a dangerous activity for humans, animals in the wild put themselves at risk just to drink water. For example, when zebras gather around a water hole, they are not "on guard" or able to fully watch their surroundings for predators. Due to this, some zebras may stay off to the side of the water hole while others drink. If one of the zebras on watch makes a warning call, the other zebras at the watering hole will immediately run away. What are the proximate and ultimate causes for this behavior in zebras?

➔ Questions?
➔ Textbook chapters/pages to review

Think, Pair, Share

Listen to the audio clip of the house finch. While you listen, try to answer this question: how do finches know to sing this unique song (i.e. how is this behavior formed)?

Types of Behavior

Behavior can be ________________________ or _________________________.

Innate behaviors:

Learned behaviors:

Note: while behaviors may lean one way more, many behaviors have both innate and learned components.

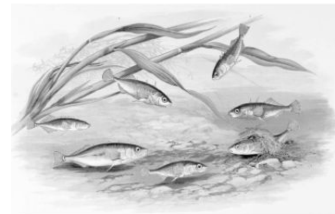
- Questions?
- Textbook chapters/pages to review

Practice FRQ

Referring back to the house finch example, design an experiment that would test whether their song is an innate or learned behavior. Make sure to identify: a) Hypothesis (null and one alternative) b) Independent and dependent variables c) Control(s) d) Constants e) Experimental design f) Results that would support or refute the hypothesis g) A justification as to why the results would support or refute the hypothesis.

Innate Behaviors

Fixed action patterns (FAPs):



Actions are ________________________

Carried out to ________________________

Triggered by a __________________________________ (external cue)

Example: stickleback fish

Male stickleback fish have bright, red bellies, unlike their female counterparts who have silver bellies. Male stickleback fish are highly territorial and will attack other males who enter their territory. When male sticklebacks see the color red (their sign stimulus) they become aggressive. In fact, it doesn't even have to be another male fish, anytime they see the color red, they become highly irritated, aggressive, and attack.

Migration:

Triggered by ____

Signal:

Examples: visual, auditory, tactile, electrical,chemical

Pheromones:

Stimulus response chains:

Seen in animal courtships

Body movement

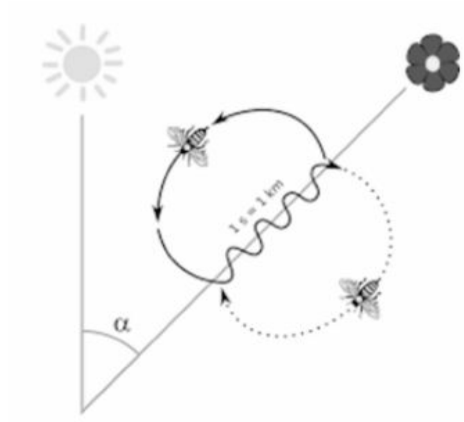
Example:

➔ Questions?

➔ Textbook chapters/pages to review

Example of a body movement signal: honey bees

In the early 1900s, it was discovered that honeybees communicate through body movements that resemble dancing. Today, it is known as the waggle dance. Worker bees will gather around a bee that has returned to the hive after foraging. The returned bee will shake its abdomen and form a figure-eight shape to tell other bees the direction where food can be found.



Directed movements:

Kinesis:

Taxis:

Phototaxis:

Chemotaxis:

Geotaxis:

Critically Think

What are possible **proximal** and **ultimate** causes for a fixed pattern action in male stickleback fish?

➔ Questions?
➔ Textbook chapters/pages to review

Learned Behaviors

Learning:

Imprinting:

Happens during:

Example: ducks following their mother

Spatial learning:

Some animals form a ________________________ map or use ______________________ as environmental cues.

Example: birds finding their hidden nests

Associative learning:

Example: associating monarch butterflies with a foul taste

Social learning:

Example: chimps breaking open oil palm nuts

Critically Think

What are possible **proximal** and **ultimate** causes for imprinting in ducklings?

➔ Questions?
➔ Textbook chapters/pages to review

The Benefit of Schools: A Critical Look at Collective Behavior in Fish

A school is a group of fish swimming in a synchronized manner. Social behavior, or the interactions that animals have with other members of their species, can have a profound effect on the individual's survival and reproductive success. In the case of many species of fish, schooling behavior is one of the most widespread forms of social behavior.

Lens of Time: Secrets of Schooling

Scan the QR code to the right, or go to the link below to watch the video. Use this video to answer the questions 1-5 below. https://www.youtube.com/watch?v=Y-5ffl5_7AI

1. Describe the work that is being done to understand schooling behavior at the University of Konstanz, Germany.

2. What is one of the leading questions regarding collective behavior that researchers are trying to answer?

3. Why is studying collective behavior, in general, quite difficult to do?

4. Currently, schooling behavior is studied using small groups of fish in nature, but what do scientists hope for in the future?

5. What is one possible proximate and ultimate cause of schooling behavior in fish?

Skipping School

Dr. Katie Peichel is an evolutionary biologist who has dedicated much of her life to studying schooling behavior in two forms of stickleback fish- one that lives in the ocean and one that lives in rivers and streams. These habitats are strikingly different and, as one might expect, differences in habitat affect behavior. Yet surprisingly, these fish are defined as the same species. Despite being the same species, these fish are given identifying names: marine sticklebacks (ocean dwelling) and benthic sticklebacks (freshwater dwelling). Marine sticklebacks have a strong tendency to school in their natural habitat, while benthic sticklebacks show less tendency to school in their natural habitat. To fully understand behavioral evolution and the differences we see within members of the same species, it is necessary to identify the genetic mechanisms that mediate behavioral change in a natural context. Genetic analysis of behavior can also reveal associations between behavior and morphological phenotypes, providing insight into the proximate mechanisms that control behavior. While this may sound easier said than done, behaviors arise from complex interactions between multiple genes and the environment, and additionally, it is difficult to pinpoint mutations that may cause natural variation in behavior between members of the same species. Despite all of this, Dr. Peichel and her team undertook a seven-year project dedicated to finding a link between schooling behavior and genetics.

1. Scientifically speaking, what does it mean that the marine and benthic stickleback fish are the same species?

2. Is it possible that the marine and benthic stickleback fish will become separate species in the future? Why or why not?

3. In response to question 3, what mode of speciation would be most likely if they were to diverge into two separate species? What type of pre or post zygotic barrier would potentially lead to the speciation event?

4. In your own words, why is it necessary to identify the genetic mechanisms that mediate behavioral change in a natural context in order to fully understand behavioral evolution?

Robo-Fish

To begin their work, Dr. Peichel and her team sought to understand the differences in behaviors between the marine and benthic stickleback fish. Specifically, she and her team first wanted to study schooling behavior in individual stickleback fish. To do this, Dr. Peichel created a robotic school of model sticklebacks. The robotic school consisted of eight fake fish (modeled to look like stickleback fish) that were attached to a robotic arm. This robotic arm was placed in a tank, and both the marine and benthic stickleback fish were exposed to this stimulus so that their behavior could be observed. Follow the link, or scan the QR code, to observe the behavior of the stickleback fish in response to the robotic fish stimulus. https://www.youtube.com/watch?v=DoLcDm64rW0

5. What differences in behavior do you observe between the marine and benthic stickleback fish?

6. Upon analyzing this data (seen in video), Dr. Peichel created a figure to depict body orientation while in the school (*Figure 1*). Proper body orientation is key in order to maintain direction, speed, and timing with the other fish in a school. What do you notice regarding body position between the wild type marine and wild type benthic stickleback fish?

7. How does body position affect the fish?

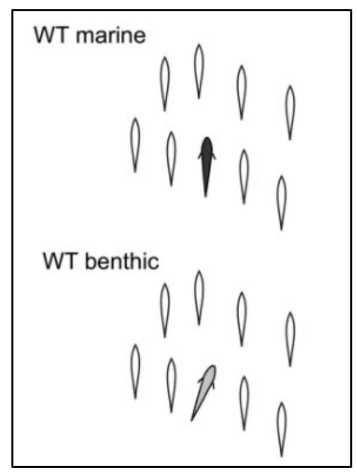
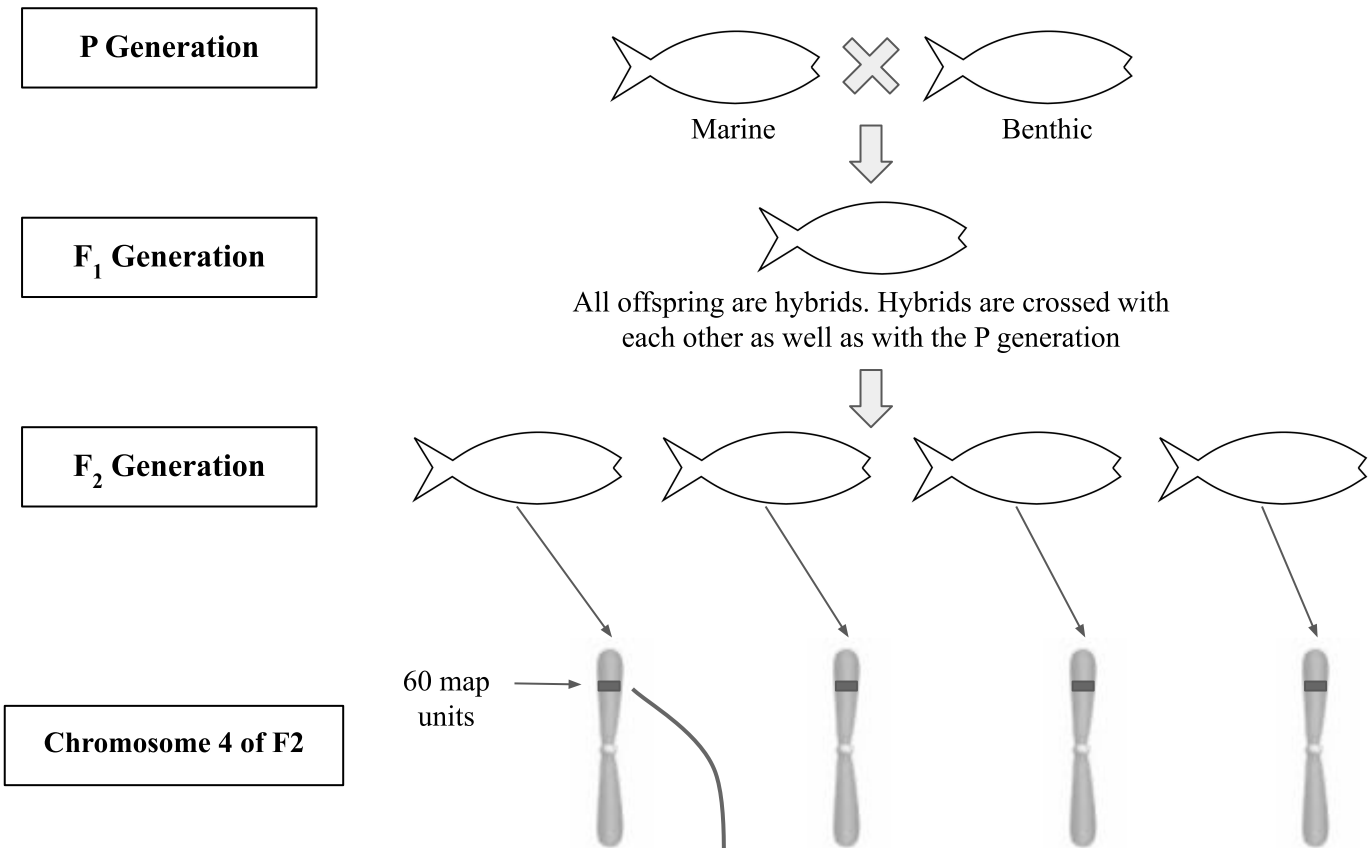


Figure 1. Wild type marine and benthic fish behavior. Open shapes represent the robotic fish.

8. Brainstorm two reasons these fish might exhibit differences in body position. Justify each reason given.

Quantitative Trait Locus Analysis

To find the link between schooling behavior and genes, Dr. Peichel and her team performed Quantitative trait locus (QTL) analysis. QTL is a statistical method that links two types of information: phenotypic data and genotypic data, in an attempt to explain the genetic basis of variation in complex traits. In order to perform a QTL analysis researchers need two strains of an organism (in this case marine and benthic sticklebacks) and a genetic marker that will distinguish the parental lines (in this case body position). To perform a QTL analysis the parental strains are crossed, resulting in heterozygous (F1) individuals. Next, these F1 individuals are then crossed with other F1 individuals or with parents, creating F2 individuals. Finally, the phenotypes and genotypes of the F2 population are sequenced. The entire stickleback genome was sequenced, so the researches had a clear idea of the number and type of genes present. Next, they examined the genes on each chromosome to look at molecular markers relating to the body position phenotype. Molecular markers that are genetically linked to a QTL influencing the trait of interest will segregate, or show up, more frequently. From this the scientists can infer that somewhere near that molecular marker must be a gene that controls that trait. The results of their QTL analysis can be seen below.



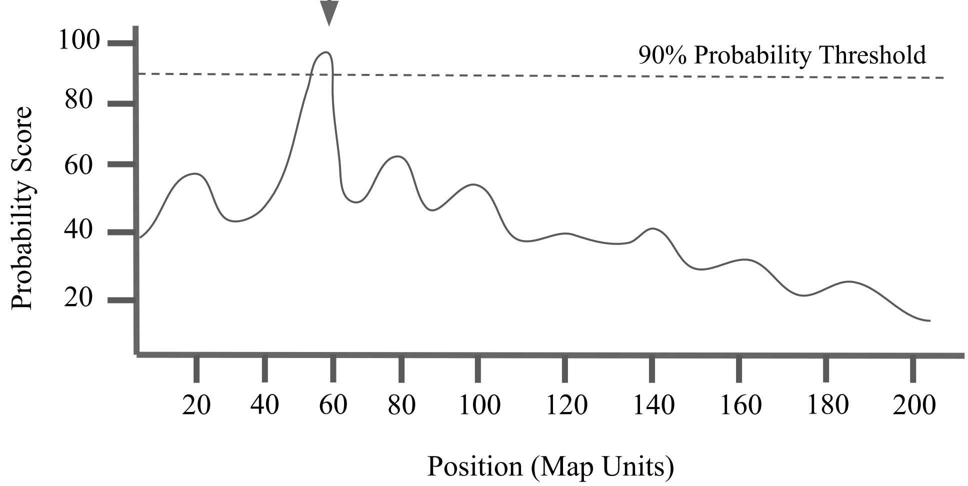


Figure 2. The x-axis represents the location of the molecular marker. The y-axis represents the probability score. Every value on the graph represents the probability that the molecular marker is highly correlated to the trait of interest. For example, at 100 map units there is a 60% chance it is near genes that correlate to body position.

9. What is the purpose of QTL analysis?

10. What are the steps in performing a QTL analysis?

11. Which chromosome contained a possible gene of interest?

12. Where is the gene of interest located (map units)?

13. Why were the F2 generation analyzed and not the F1?

14. What is the purpose of Figure 2?

15. What was the purpose of using molecular markers?

The Eda Gene

By analyzing the data from the QTL analysis, Dr. Peichel and her team were able to determine that the gene associated with the molecular marker was the *Ectodysplasin (Eda)* gene. The *Eda* gene produces the Eda protein that controls the formation of bony plates on the sides of the sticklebacks. These bony plates act as "armor" for the fish and aid in protection against predation. It is now known that the *Eda* gene plays many other important roles in the fish, like affecting schooling behavior and modulating the immune system of the fish; both of these mechanisms are still not fully understood, but links to each have been established. The benthic sticklebacks have fewer bony plates, meaning they have less Eda protein in their bodies. In comparison to the marine stickleback fish, which have more bony plates, meaning they have more Eda protein. This gene is also found in other organisms and controls the formation of teeth and hair.

16. In order to form bony plates, the stickleback fish need a high amount of minerals. Knowing this, why might the benthic fish have adapted over time to produce less bony plates?

17. Predict what would happen to benthic stickleback fish behavior if they were genetically modified such that transcription and translation of the *Eda* gene were promoted?

18. Predict what would happen to marine stickleback fish if the *Eda* gene were knocked out.

19. Generally speaking, how might the benthic stickleback fish still be able to survive and reproduce even through they do not produce as much Eda protein?

20. Think about the topography of the ocean versus the topography of rivers/streams. Why might the marine stickleback school, while the benthic stickleback do not?

Natural Selection

Natural selection favors:

Foraging:

Animals better at foraging will be more ______________________in finding food

Mating behaviors:

_____________________________________ can result from sexual selection

________________________ behaviors tend to increase fitness

Examples: predator warnings, kin selection, pack behavior

Altruism:

Example: naked mole rat colonies have only one reproducing female (queen), who will only mate with a few males (kings). The other nonreproductive members will sacrifice themselves to protect their queen and kings.

Responses in Plants

Since _________________________ is critical to a plant's survival, plants can respond to ____________.

Phototropism:

Photoperiodism:

➔ Questions?
➔ Textbook chapters/pages to review

Plants also have mechanisms of defending themselves against herbivory:

Physical defenses:

Chemical defenses:

Example: lima bean plants that are being damaged release volatile chemicals that surrounding lima bean plants can sense, causing them to release compounds making themselves less susceptible to herbivory

_________________ composition can affect plants.

The pH of soil can affect ____ in some plants

Example: hydrangea blooms turn different colors based upon soil pH (blue-pH 5, pink- pH 7)

Practice FRQ

Walking in the woods, a researcher encountered a group of ravens feeding on the carcass of a moose. The birds were being unusually loud. The researcher realized that the loud calls of the ravens were attracting more ravens to the area, so they could also feed. This is a striking contrast to other species, where food is defended. Discuss the possible proximal and ultimate causes for the raven's behavior while feeding. Identify one reason why other species may defend their food.

➔ Questions?
➔ Textbook chapters/pages to review

Review: Behaviors

1. What is meant by the phrase “nature vs nurture?”

2. Of “nature” and “nurture, which do you think has more of an affect on behavior? Provide reasoning for your thoughts.

3. Design an experiment that would test the genetic component of behavior.

4. Differentiate between an innate behavior and a learned behavior. Provide an example of each.

5. Describe a proximate cause. How does it differ from an ultimate cause?

6. Describe the difference between kinesis, taxis, and migration.

7. Research the Greylag goose egg-retrieval behavior. Describe the behavior and identify the type of innate behavior it represents?

8. Identify the sign stimulus in the Greylag goose egg-retrieval.

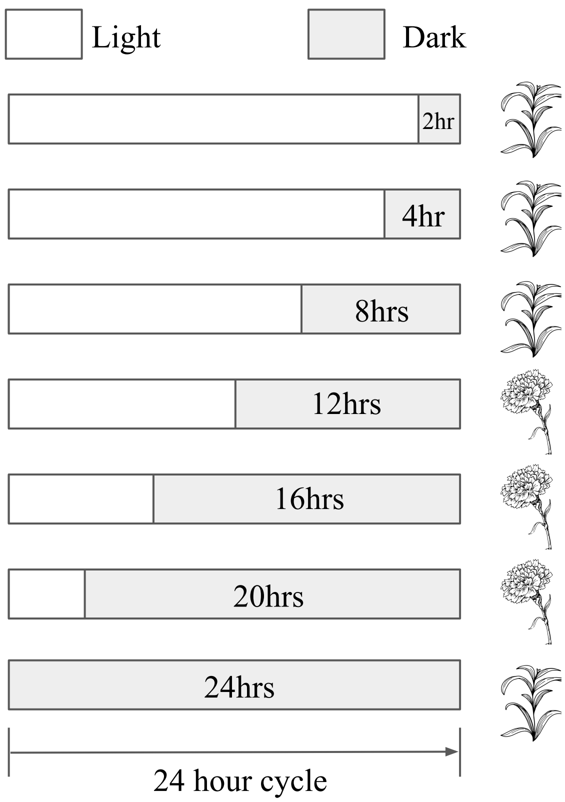
9. Let’s say you put a golf ball (which looks roughly like an egg) by a nest of a Greylag goose. What would happen?

10. Some types of mosquitoes are known to use landmarks as a mechanism to find their nesting sites. If a researcher were to move the landmark away from the nest (but the nest remains unmoved in the same location), would the mosquito still be able to find the nesting site? Why or why not?

11. Why does altruism exist in nature? How can you explain altruistic behaviors between different species? For example, some species of birds will help other birds nest and forage for food.

Photoperiodism: A Plant's Response to Light

Plants have critical events that occur at different times of the year. For example, seed germination, flowering, and breaking bud dormancy. But how do plants know when it is the right time to go through these critical stages of their life cycle? A researcher decided to investigate this by examining the *Xanthium strumarium* plant. The researcher proposed that the length of daylight is responsible for this plant's life cycle. She designed an experiment to test her hypothesis. Examine the experimental set up below.



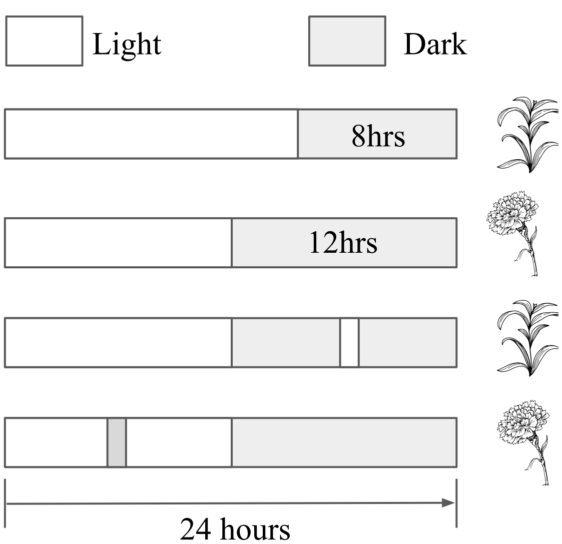
1. State a possible null and alternative hypothesis for the experiment.

2. What are the independent and dependent variables?

3. What preliminary conclusions can you draw from the results of this experiment?

The researcher presented her findings to her colleagues. They noted that this experiment failed to provide evidence to support the length of daylight as the critical period responsible for the plant knowing when to flower. What if it was length of night, or the absence of light, that was critical for the flowering of the plant? With this information, she revised her experiment by exposing the plants to a flash of light during the periods of darkness, and a flash of darkness during periods of light.

4. What are the controls in this experiment?



5. What is the critical period for the plant, the period of light or the period of darkness? Justify your response by referring to the data.

Next, the researcher wanted to determine if the timing of the flash of light, whether it was early or late into the dark period, had an affect on the flowering plants. The researcher exposed *Xanthium strumarium* plants to a cycle of 12 hours light/12 hours dark and exposed the plants to a 10 minute burst of light at different times within the dark period. She recorded the percent of flowering plants post exposure. The data is in the chart below.

Time of burst of light (from the beginning of dark period)	% of plants flowering
0	100
2	86
4	35
6	5
8	41
10	78
12	95

6. Graph the data in the table using the space provided above.

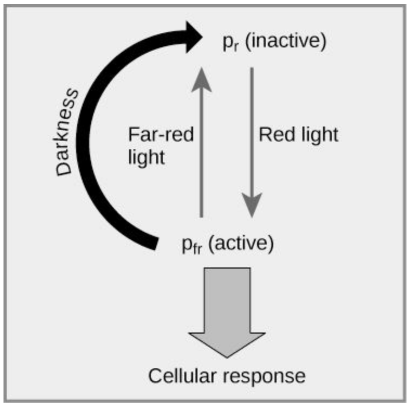
7. What conclusions can you draw from this data?

8. At what time was the exposure to light the most detrimental to the flowering of the plant?

9. Hypothesize why light exposure at some times of the dark period affect the plant <u>more</u> than at other times of the dark period?

Phytochromes

It is now known that plants use a photoreceptor protein, called phytochrome, to sense changes in the length of light/dark periods. Since the length of daylight changes throughout the year, this is how plants determine seasons, and when it is appropriate to flower. During the day, sunlight (red light) triggers photosynthesis, which quickly converts phytochrome to its active form, P_{fr}. At night, far-red light slowly converts P_{fr} back into its inactive form, P_r.

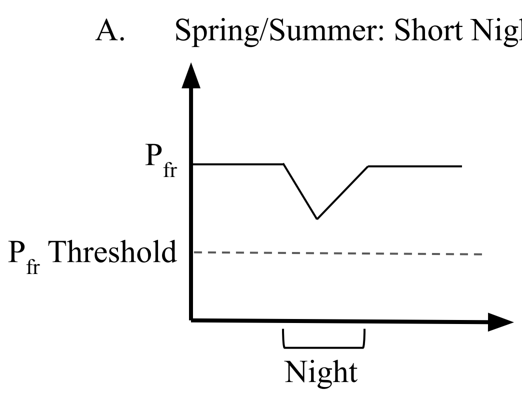


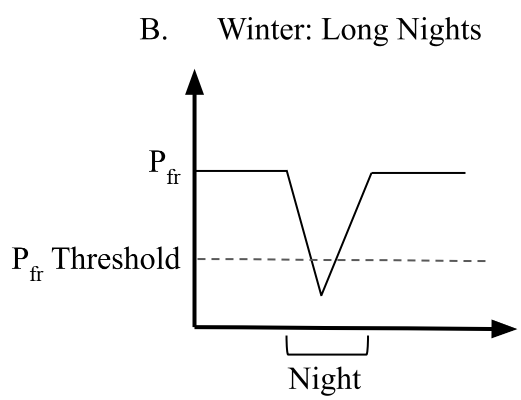
Short day vs Long Day Plants

Plants can be classified as short-day (long night), long-day (short night), or neutral-day plants based upon their critical period of darkness needed to flower. The naming (short-day, long-day) has to do with the fact that critical night periods were discovered after the naming; scientists originally believed that light regulated flowering seasons. Short-day plants require 12 hours or more of darkness per day to flower, while long day plants require less than 12 hours of darkness per day to flower. In short-day plants, P_{fr} acts as an inhibitor to flowering. In long-day plants, P_{fr} promotes flowering.

10. Looking back at the data for *Xanthium strumarium*, would you say this plant is a short-day plant or a long-day plant? Why?

Examine the graphs below showing the relative phytochrome levels for short-day plants in the spring/summer season versus the winter season.





11. Which graph (A or B) represents the season when a short-day plant would flower?

12. Using the graph selected in the previous question, explain how P_{fr} regulates flowering in short-day plants.

13. Knowing this, now explain why short bursts of light during the dark period caused the *Xanthium strumarium* plant to not grow.

Topic 2

Ecosystems and Energy

Ecosystem:

Biotic factors:

Abiotic factors:

Practice Time

Look outside and list three biotic factors and three abiotic factors. List them below:

Think, pair, share: what are the first two laws of thermodynamics?

1st law:

Law of conservation of mass:

2nd law:

A net gain in energy results in:

A net loss of energy results in:

- Questions?
- Textbook chapters/pages to review

Metabolic Rate

Metabolic rate:

Can be measured in:

________________ is used in cellular respiration and ______ is produced as a by-product

An animal's metabolic rate is related to its:

Smaller organisms =

Larger organisms =

Why might this be?

Ecosystems and Energy

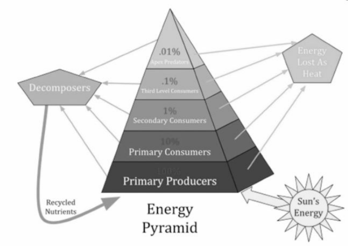
Organisms use different strategies to regulate body temperature.

Endotherms:

Ectotherms:

➔ Questions?
➔ Textbook chapters/pages to review

Trophic Levels



Species can be grouped into trophic levels based upon their main source of:

Unlike mass, ______________________ CANNOT be recycled.

The ____________ constantly supplies energy to ecosystems.

Primary producers (autotrophs) use ___________________________ to synthesize organic compounds

Some organisms are ____________________________ (vs photosynthetic) meaning they produce

food using the energy created by ____________________________________

Heterotrophs:

Primary consumers:

Secondary consumers:

Tertiary consumers:

Decomposers:

Include fungi and many prokaryotes

Important for recycling chemical elements

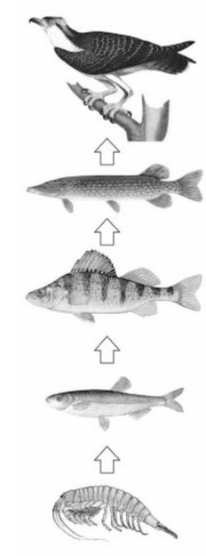
➔ Questions?
➔ Textbook chapters/pages to review

Trophic Structure

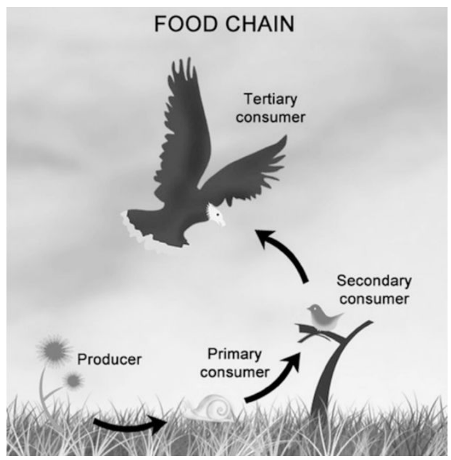
The trophic structures of a community are determined by the ______________________________________

between organisms

Food chain:



Food webs:



Notice the arrows show the transfer of energy.

Trophic Levels

Any changes to the availability of ______________________ can disrupt ecosystems

For example:

- If energy resources change, so can the __________________ and ________________ of trophic levels (Increase energy, increase trophic levels/size; decrease energy, decrease trophic levels/size)
- A change at the ________________________ level can affect the number and size of the remaining __________________ levels

➔ Questions?
➔ Textbook chapters/pages to review

Activity: Create Your Own Food Web

Objective: Now that we have learned about trophic levels, you will be tasked with creating your own food web to demonstrate your understanding of the subject matter.

Directions: Your teacher will assign you <u>one</u> animal that MUST go into your food web. Your task is to research the predators and prey of this animal to create the beginning of a food web. Once you have researched the predators and prey for your animal, research the predators and prey for the other animals that are now included in your food web. Continue this process until you have at least 10 organisms in your food web. You must include producers, primary consumers, secondary consumers, and tertiary consumers. You will create your completed food web on the next page. Once you complete your food web you will write a short analysis discussing the number of links between trophic levels can contribute to the overall stability of the community.

Checklist:

- ❏ At least 10 organisms (*1 pt. per organism; 10 pts. total*)
- ❏ Each trophic level is labeled (*4 pts. total*)
    - ❏ Includes a minimum of one of each: producers, primary consumers, secondary consumers, tertiary consumers
- ❏ Arrows indicating the flow of energy (*1 pt. total*)
- ❏ A picture of each organism included (*1 pt. total*)
- ❏ A discussion on the link between trophic levels and a community's stability (*4 pts. total*)

List: Your teacher will assign you **one** organism from the list below.

- Red-lipped batfish (*Ogcocephalus darwini*)
- Praying mantis (*Mantodea*)
- Venus fly trap (*Dionaea muscipula*)
- Panda ant (*Euspinolia militaris*)
- Blue sea dragon (*Glaucus Atlanticus*)
- Pink fairy armadillo (*Chlamyphorus truncatus*)
- Bush baby (*Galagidae*)
- Mantis shrimp (*Stomatopoda*)
- Snowy owl (*Bubo scandiacus*)
- Wrinkle-faced bat (*Centurio senex*)
- Narwhal (*Monodon monoceros*)
- Sea pig (*Scotoplanes*)
- Fennec fox (*Vulpes zerda*)
- Elephant shrew (*Macroscelididae*)
- Mary river turtle (*Elusor macrurus*)
- Leafy seadragon (*Phycodurus eques*)
- Bush viper (*Atheris*)
- Sloth (*Folivora*)

Total Points: _______/20 points

Completed Food Web

Construct your food web in the space below.

Analysis

Use the space below to write a short analysis of your food web. Include a discussion on how the number of links between trophic levels can contribute to the overall stability of the community. If any animal in your food web is endangered, discuss what could happen to the stability of the food web if that organism were to be removed.

Primary Production

Primary production:

_____________________ producers set a “spending limit” for the entire ecosystems energy budget

Gross primary production (GPP):

Net primary production (NPP):

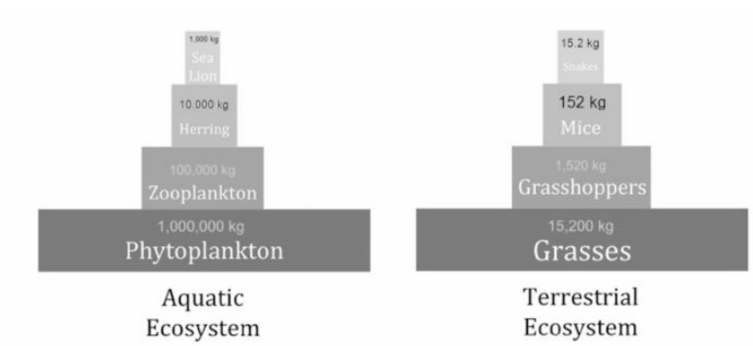
Satellite images show that different ecosystems have varying NPP

- What areas have higher NPP and why?

Secondary Production

Secondary production:

The transfer of energy between trophic levels is at around ________ efficiency



→ Questions?
→ Textbook chapters/pages to review

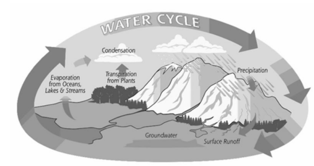
Matter Cycling

Unlike energy, matter cycles through ecosystems

Biogeochemical cycles:

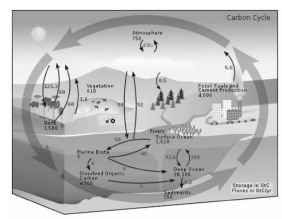
Water Cycle

Biological importance:



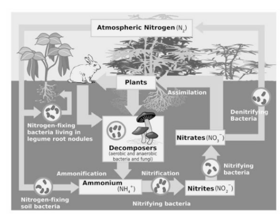
Carbon Cycle

Biological importance:



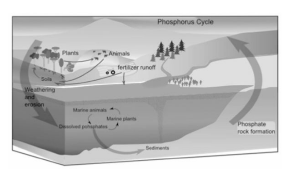
Nitrogen Cycle

Biological importance:



Phosphorus Cycle

Biological importance:



- ➔ Questions?
- ➔ Textbook chapters/pages to review

Populations

Population:

Population ecology:

Density:

Can be determined by:

____________________ the number of individuals in the population (rarely done)

____________________ techniques (count small areas, average the areas, and then use the averages to estimate total population size)

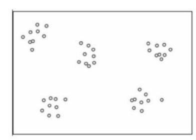
Knowing a population's density provides more information about its relationship to the _______________________ it uses

Critically Think

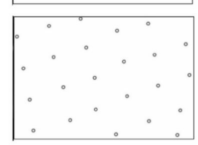
Species with large body sizes generally have lower population densities than species with smaller bodies. Why might this be?

Dispersion:

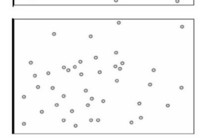
Clumped:



Uniform:



Random:



The size of a population is not ______________, affected by:

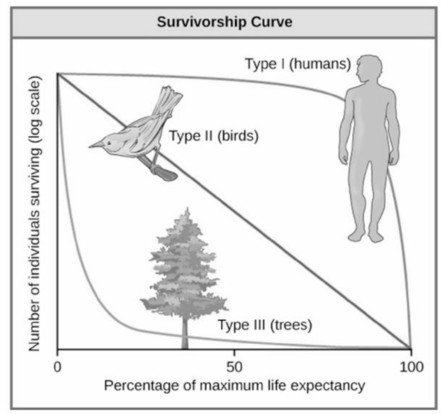
➔ Questions?
➔ Textbook chapters/pages to review

Demography:

Life table:

Represented by a ________________________________ curve

Type I curve:



Type II curve:

Type III curve:

Change in Population Size

The per capita rate of increase can be calculated using this formula:

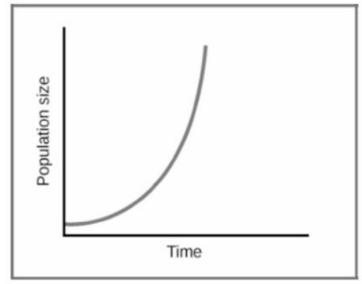
$$\frac{dN}{dt} = B - D$$

- ➔ Questions?
- ➔ Textbook chapters/pages to review

Growth Models

There are two models for population growth

Exponential growth model:



A population growing exponentially grows at a ________________________ rate.

- ______ shaped curve
- Calculated using the formula:

$$\frac{dN}{dt} = r_{max} N$$

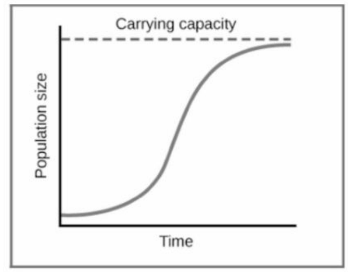
Practice Problem

A population of bunnies is growing exponentially. The growth rate of the population, r, is 1.5 and the current population size, N, is 3,000 individuals. How many bunnies are being added to the population each year?

➔ Questions?
➔ Textbook chapters/pages to review

Logistic Growth

Logistic growth model:



- The density of individuals exceeds the system's ________________________ availability.
- Calculated using the following formula:

$$\frac{dN}{dt} = r_{max}N\left(\frac{K-N}{K}\right)$$

Practice Problem

A hypothetical population has a carrying capacity of 2,000 individuals and r_{max} is 1.0. What is the population growth rate for a population with a size of 1,200 individuals? What is happening to this population?

➔ Questions?
➔ Textbook chapters/pages to review

Population Dynamics

Populations are influenced by ______________________________________ and

_________________________ factors.

Life history:

Three variables affect life history:

1. ___________ reproduction begins
2. ___________________ the organism can reproduce
3. The ___________________ of offspring produced per reproductive episode

K-selection (density-dependent selection):

Seen in __________ density populations that are close to ______________________________ (K)

R-selected (density-independent selection):

Seen in __________ density populations with little _________________________.

Density-dependent regulation:

Competition, predation, toxic wastes, territoriality, disease, intrinsic factors (ie reproduction rates)

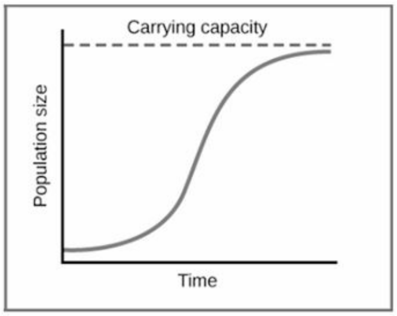
Density-independent regulation:

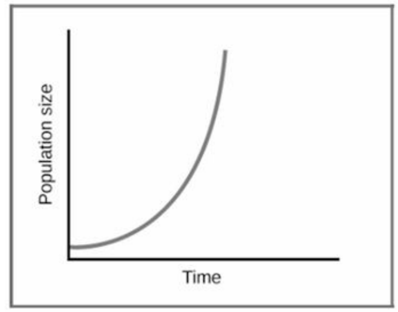
Weather, climate, natural disasters

➔ Questions?
➔ Textbook chapters/pages to review

Review: Growth Models

1. Label the graphs below as the growth model they represent:





2. Discuss reasons why the exponential growth model is not sustainable.

3. Explain the concept of carrying capacity (K).

4. What is the relationship between population size (N), carrying capacity (K), and growth rate (r) in the logistic growth model?.

5. What is the difference between K-selected and r-selected populations?

Practice Problems

Rate	**Population Growth**	**Exponential Growth**	**Logistic Growth**
$\frac{dY}{dt}$	$\frac{dN}{dt} = B - D$	$\frac{dN}{dt} = r_{max}N$	$\frac{dN}{dt} = r_{max}N\left(\frac{K-N}{K}\right)$

dY = amount of change

dt = change in time

B = birth rate

D = death rate

N = population size

K = carrying capacity

r_{max} = maximum per capita growth rate of population

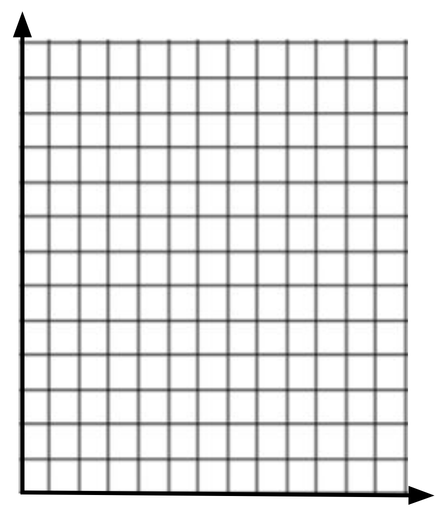
1. Suppose in 2015 a population of 500 squirrels lived in a chaparral region of Southern California. If every year 55 squirrels were born and 32 squirrels died, calculate and interpret the following:

   a. The population growth rate

   b. The per capita growth rate of the squirrels over a year

   c. Using the data calculated above, complete the chart below and graph the data using the space given

Year	Population
2015	
2016	
2017	
2018	
2019	



2. A hypothetical population has a carrying capacity of 1,500 individuals and r_{max} is 1.0. What is the population growth rate for a population with a size of 1,800 individuals? What is happening to this population?

3. A region of North America has a population of 65,000 deer in 2016. The deer are kept in check by a wolf population. The carrying capacity of this region allows for 82,000 deer. For this example, assume the r_{max} is 1.0. With this information, answer/calculate the following:

a. Is the population above or below carrying capacity?

b. What will the population growth rate be for 2016?

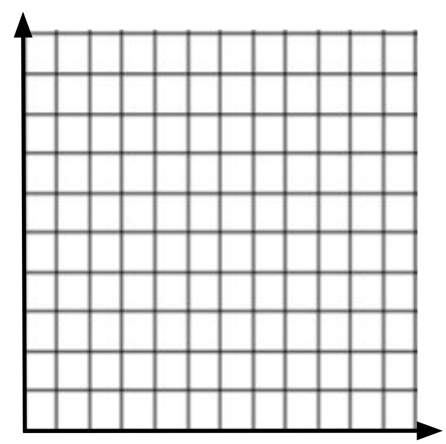
c. What will the population size at the start of 2017 be?

d. Perform calculations to fill in the chart below:

Year	Population	Population Growth Rate
2016		
2017		
2018		
2019		

e. Graph the data calculated in part (d).

f. What happens to the population size from 2016-2019?



g. If something happened in North America to decrease the wolf population, how would this affect the deer population? How would the graph from (e) change?

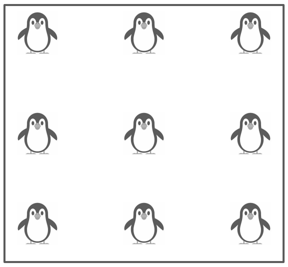
Let's explore what happens to the size of a population when it overshoots its carrying capacity. Sometimes, for a brief period of time, populations can exceed their carrying capacity. This can happen, for example, if food becomes a limiting factor; there may be a delay before reproduction declines, causing the population to exceed its carrying capacity briefly.

1. Assume that r_{max} is 1.0 and the carrying capacity is 2,000. Calculate the population growth rate when N= 2,100 and when N= 2,500 individuals.

2. If r_{max} is doubled, <u>predict</u> how the population growth rate will change for the two populations given in the previous problem. Then, <u>calculate</u> the population growth rate for the same two cases above, except this time assuming r_{max} is 2.0.

3. Sometimes after a population briefly exceeds their carrying capacity, there will be a drop in population numbers well below their carrying capacity before it begins increasing again. What is a possible explanation for the drop seen after populations exceed their carrying capacity?

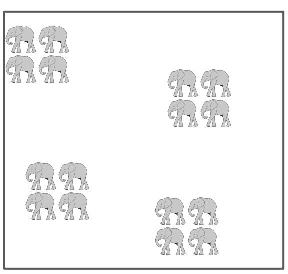
Population Ecology Practice Problems: Part 1



The image to the left shows the dispersion pattern of king penguins. King penguins are monogamous animals and will stay with their mate for all of mating season. During mating season, a female will lay an egg. The incubation period for the egg is approximately 54 days. During this time, the mates will take turns incubating the egg a week at a time by holding it between their feet. While one mate incubates the egg, the other searches for food. On average, only pairs that reproduce during the first half of the mating period are successful in fledging a chick because they are more likely to have claimed territory and found food early in the season.

1. Identify the dispersion pattern and discuss the possible reasons for the dispersion pattern seen in the king penguins during nesting season.

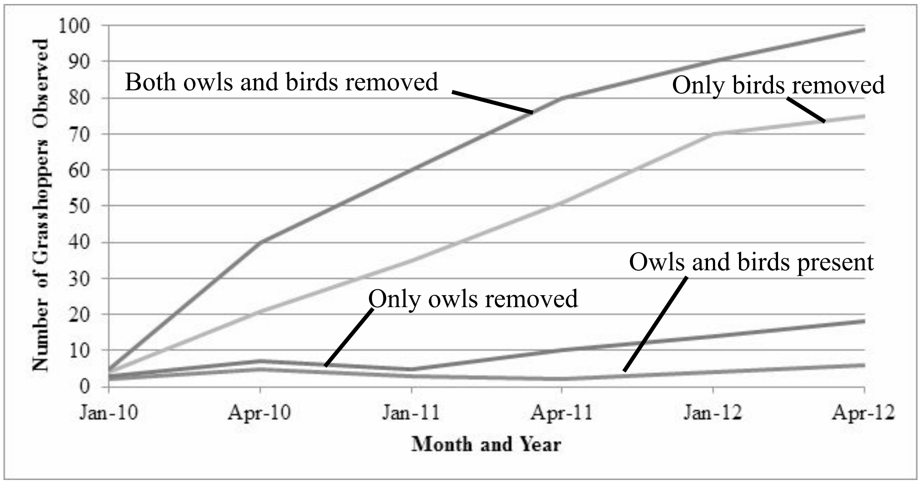
2. Identify and describe one density-dependent factor and one density-independent factor that could lead to changes in the king penguin population.



The image to the left shows the dispersal pattern of elephants in sub-Saharan Africa. Elephants are herd animals with a definite social structure. Herds are lead by the matriarch, usually the oldest female. The rest of the herd consists primarily of females, and their focus is successfully raising offspring. Males leave the herds during adolescence to lead a more solitary life, or form small herds of their own with other males. Males are usually not too far off from their offspring though, and can watch their offspring grow from a far.

3. Identify the dispersion pattern and discuss the possible reasons for the dispersion pattern seen in the sub-Saharan elephants.

4. Due to poaching, the African elephant population has been declining rapidly. Discuss the effects that this decline in numbers could have on the surrounding ecosystem.



An ecology professor was interested in studying the relationships between owls, birds, and grasshoppers in a local field by his university. He conducted a two year study that focused on determining if owls and birds were limiting biotic factors to the grasshopper population. Both owls and birds eat grasshoppers, but he hypothesized that birds were a more significant limiting factor than owls.

1. What was the positive control in the experiment?

2. What experimental group lead to the greatest increase in the grasshopper population?

3. If removing owls did not significantly increase the grasshopper population in comparison to the control, propose a reason why removing both owls and birds lead to a greater increase in the grasshopper population than removing birds only.

4. From this experiments, what conclusions can be drawn about the limiting biotic factors affecting the grasshopper population?

5. If the researcher wanted to further his study by examining possible limiting abiotic factors to the grasshopper population, what could he include? List 3 possible abiotic factors.

6. From the abiotic factors listed above, choose one and create a hypothesis that he could test.

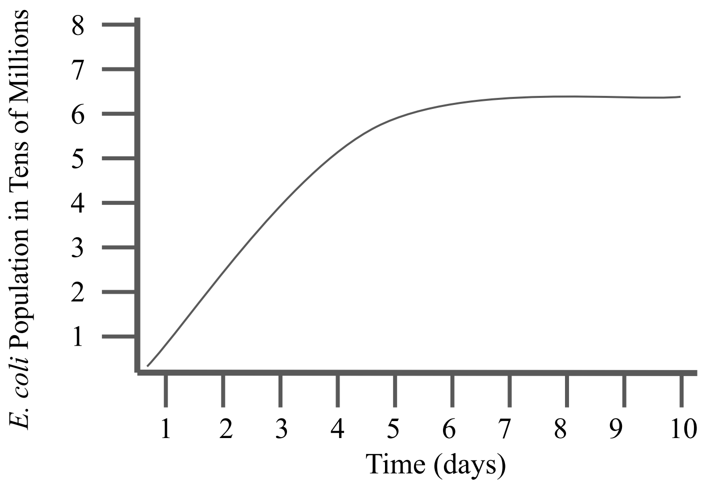
Population Ecology Practice Problems: Part 2

Rate	**Population Growth**	**Exponential Growth**	**Logistic Growth**
$\frac{dY}{dt}$	$\frac{dN}{dt} = B - D$	$\frac{dN}{dt} = r_{max} N$	$\frac{dN}{dt} = r_{max} N \left(\frac{K - N}{K} \right)$

dY = amount of change
dt = change in time
B = birth rate
D = death rate
N = population size
K = carrying capacity
r_{max} = maximum per capita growth rate of population

Directions: Read each problem and show your work in the area provided. Be sure to box all final answers.

1. A scientist is conducting a survey on the variety and number of organisms located around the science building, which sits on a total of 10 acres. On the first day of his survey he counted a total of 640 squirrels. What is the density of the squirrels per acre in the surveyed area? *Round your answer to the nearest whole number.*	Show work here:
2. On the second day of the survey, the scientist counts 200 sage plants. He makes note of the fact that the sage plants exhibit uniform distribution across the 10 acre campus. What would the population size of the sage be if the sage grew in an area that was 20 acres? *Round your answer to the nearest whole number.*	Show work here:
3. On the third day of the survey the scientist counts 352 crows. The scientist has teacher assistants (TAs) helping him with his research project. He tasks the TAs with tagging all of the crows in the region; to tag the crows the TAs place a small band with an ID # around each crow's right ankle (this causes no harm to the crows). Over the course of a month, the TAs record the number of births and deaths in the crow population. They find that 3 crows died and 10 crows were born. What is the population size of the crows at the end of the month?	Show work here:
4. The TAs repeat this process of tagging birds, except this time they tag a population of 522 blue jays on the fourth day of the survey. Over the course of the study they calculate the per capita birth rate to be 0.10 and the per capita death rate to be 0.07. With this information, calculate the following: a. How many blue jays were born during the study? *Round your answer to the nearest whole number.* b. How many blue jays died during the study? *Round your answer to the nearest whole number.* c. What is the per capita growth rate of the blue jay population? *Round your answer to the nearest hundredth.*	Show work here:

5. As the scientist and his team of TAs are performing their survey, they are able to make many observations. Identify each observation below with the correct type of population regulation: density dependent or density independent regulation. a. Ravens are unable to nest on campus because the crows will attack them. b. An exceptionally rainy season created some flooding on campus, which wiped out the cricket population in that region.. c. High squirrel population density caused female squirrels to inhabit poor quality territories, which reduced their reproductive success.	Show work here:
6. While the researcher and TAs were outside, microbiologists were inside of the science building performing experiments using *E. coli*. Examine the graph below to answer the questions.  a. Calculate the mean rate of growth from day 2 to day 4. b. What is the carrying capacity? c. How would this growth curve for *E. coli* be affected if it were cultured with a competing bacteria?	Show work here:
7. After an extensive survey of habitat and food availability for the squirrel population of 640, the scientist concluded that they have a carrying capacity of 950 squirrels. What is the maximum population growth rate if the population grows to be 720 in one year?	Show work here:
8. Let's say in five years, the scientist comes back to analyze the squirrel population because the science campus is expanding. He finds that the squirrel population has stabilized around 945 squirrels. Using this data he calculates the r_{max} to be 0.43. With the campus expanding, more area will be available for the squirrels to inhabit. What will the population size of squirrel be one year after the campus has expanded if it increases the carrying capacity to 1200? Assume r_{max} does not change.	Show work here:

Communities

Community:

Niche

Habitat:

Ecological niche:

Fundamental niche:

Realized niche:

Interspecific Interactions

Interspecific interactions:

- Competition
- Predation
- Herbivory
- Symbiosis (parasitism, mutualism, commensalism)
- Facilitation

→ Questions?
→ Textbook chapters/pages to review

Competition

Competition:

Competitive exclusion principle:

The competitor with even a slightly better advantage will eliminate the inferior competitor

Niche partitioning:

Predation

Predation:

Adaptations of both predators and prey have been refined by:

- Cryptic coloration:
- Batesian mimicry:
- Mullerian mimicry:

Herbivory

Herbivory:

➔ Questions?
➔ Textbook chapters/pages to review

Symbiosis

Symbiosis:

Parasitism:

Mutualism:

Commensalism:

Facilitation

Facilitation:

Common in:

Species Diversity

Species diversity (biodiversity):

Species richness:

Relative abundance:

Note: Biodiversity boosts ecosystem ______________________________. The greater the biodiversity in an ecosystem, the more ________________________ it is.

➔ Questions?
➔ Textbook chapters/pages to review

Practice Problem

1. Examine the two tree communities below. Each has 100 individuals distributed among five tree species (A, B, C, D, and E).
    a. Community 1: 20A, 20B, 20C, 20D, 20E
    b. Community 2: 5A, 25B, 15C, 20D, 35E

What can you conclude about their relative species richness and abundance?

Species Diversity

Simpson's diversity index:

$$\text{Diversity Index} = 1 - \Sigma\left(\frac{n}{N}\right)^2$$

High diversity index means _____________ biodiversity; low diversity index means _____________ biodiversity.

High-diversity communities are more resistant to ____________________________________

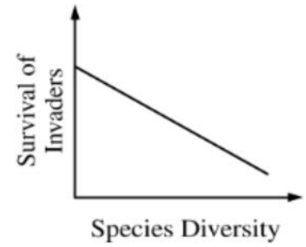
Example: a ship bringing produce from another country may have insects in the crates holding the produce

The intentional or unintentional introduction of an invasive species can allow the species to exploit a new ___________________ that is free of _________________________ or ___________________________.

➔ Questions?
➔ Textbook chapters/pages to review

Practice Problem

Examine the graph below. Describe what it represents in terms of the relative survival of invasive species in comparison to native species biodiversity.



Keystone Species

Some species play a more pivotal role than others in a community

Keystone species:

- Example: coral
    - Coral reefs serve as a keystone species because:
- Example: honey bees
    - Bees are a keystone species because:

Keystone species, producers, and essential abiotic and biotic factors contribute to maintaining the

__________________________ of the ecosystem.

➔ Questions?
➔ Textbook chapters/pages to review

Disturbances

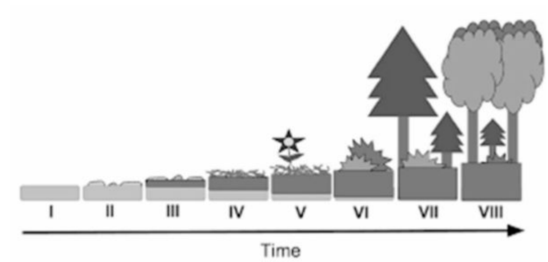
Disturbances can also influence species ________________________ and ________________________.

Disturbance:

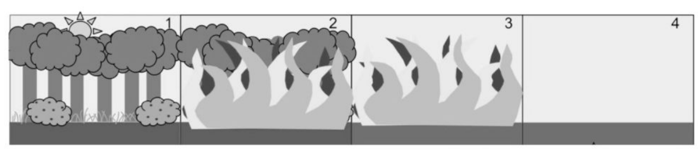
Fires, droughts, human activities, etc

Ecological succession:

Primary succession:



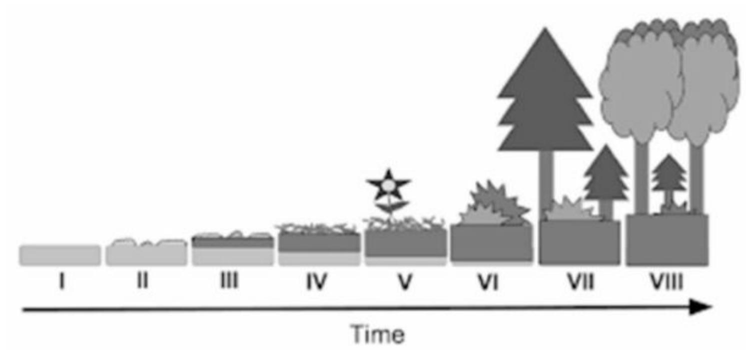
Secondary succession:



➔ Questions?
➔ Textbook chapters/pages to review

Practice Problem

Using the image below, identify two changes in abiotic conditions that would lead to the succession seen. Justify each condition identified (be specific as to whether or not the condition increased or decreased).



Human Disturbances

Human activity is the _________________________ disturbance to an ecosystem.

The main threats to biodiversity are:

- Habitat loss
- Invasive species
- Overharvesting
- Global change

Habitat loss:

- Clear cutting, cattle grazing, farmland

Invasive species:

Overharvesting:

- Harvesting of ivory in elephants (now banned)
- Overfishing

➔ Questions?
➔ Textbook chapters/pages to review

Global change:

- Air/water pollution
  - Acid rain
- CO_2 emissions
- Ocean acidification

Human disturbances have led to a significant increase in the number of ______________________ species

- Many species that are now _________________________ could potentially provide food, medicine, and fibers
- Scientists believe we are currently in a _______________________________

Biogeographical Factors

Biogeographical factors:

Latitude:

Area:

Pathogens

Pathogens:

➔ Questions?
➔ Textbook chapters/pages to review

Practice FRQ

In a certain forest community, a dominant plant species has recently been infected with a virus that disrupts the thylakoid membranes of the plants. **Identify** the process in photosynthesis that would be most affected and **explain** the most likely effects the virus would have on the individual plants.

➔ Questions?
➔ Textbook chapters/pages to review

Dry Lab: Mark-Recapture

Essential Knowledge: wildlife biologists use sampling techniques, like mark and recapture, to estimate the population size of wildlife. In this dry lab you will be utilizing the mark and recapture technique to estimate the size of a population using dry beans.

Pre-Lab: *the pre-lab must be completed in your lab notebook before coming to class. Refer to the 'Lab Notebooks' section of the "How to Write a Formal Lab Report" document to make sure that you complete your lab notebook appropriately. This is a great lab to add to your lab portfolio as well!* This lab will require a formal lab report due on:

____

Pre-Lab Questions: follow the link or scan the QR code to learn more about the mark-recapture method. Use this video to answer the pre-lab questions in your lab notebook.. https://www.youtube.com/watch?v=240806aPHVg

1. Define the mark-recapture technique.
2. Why do scientists utilize the mark-recapture technique?
3. What is the basic formula used in the mark-recapture technique? Define each variable.
4. What formula can you use if you repeat the mark-recapture technique?

Materials:

- 1-gallon sized plastic or paper bag
- 1-measuring cup
- Two colors of beans that are roughly the same size and shape
    - 1 beaker of light-colored beans
    - 1 beaker of dark-colored beans

Procedure:

1. Measure 2-cups of light-colored beans from the beaker and place into your bag. Make an educated guess as to how many light-colored beans you just placed in the bag, but do NOT actually count them. Record this estimate in Table 1.
2. Take a handful of light-colored beans out of your bag and place them on your desk. This represents your <u>first</u> capture of a group of organisms (C_1). Count these beans, set them to the side, and record the number as your value for C_1 for Sampling Time 1 in Table 1.
3. You will now **"mark"** the organisms (beans) you just captured. Rather than physically marking the organism like you would in the wild, you will simply replace the light-colored beans with dark-colored beans (same as Part I). <u>The number of beans you marked now become the number of marked individuals in the population for your next sample</u>. Record this number of marked beans as the value for M_2 for Sampling Time 2 in Table 1 (it will be the same number as for C_1). *Note that M_1 for Sampling Time 1 is 0 because there were no marked individuals originally.*
4. Return the marked beans back to the population (bag) and shake well to thoroughly mix. *Place the light-colored beans back into the beaker.*
5. Without looking, grab a handful of beans from the bag and place them on your desk. This represents your <u>second</u> capture of a group of organisms. Count the total number of beans you grabbed in this handful (regardless of whether or not they are marked) and record this number as the value for C_2 in Table 1.

6. Examine the handful of beans you gathered in step 5 above:
   a. Count the number of beans that are marked. Record this number as your value for R_2 in Table 1. *Note that R_1 for sampling time 1 is 0 because there were no captured individuals originally*.
   b. Count the number of unmarked (light-colored) beans, and then **mark** them by replacing them with dark-colored beans. Place the light-colored beans back into their respective beaker.
      i. Record the number of individuals you just marked under M_2 and record the resulting sum as the value for M_3 for Sampling Time 3 in Table 1. The represents the total number of marked individuals now in the population. Return all the beans from this second collection (which are now all marked, and therefor dark-colored) to the bag. Shake thoroughly.
7. Without looking, grab a handful of beans from the bag and place them on your desk. This represents your <u>third</u> capture of a group of organisms. Count the total number of beans you grabbed in this handful (regardless of whether or not they are marked) and record this number as the value for C_3 for Sampling Time 3 in Table 1.
8. Examine the handful of beans you gathered in step 7 above:
   a. Count the number of beans that are marked. Record this number as your value for R_3 in Table 1.
   b. Count the number of unmarked (light-colored) beans, and then **mark** them by replacing them with dark-colored beans. Place the light-colored beans back into their respective beaker.
      i. Record the number of individuals you just marked under M_3 and record the resulting sum as the value for M_4 for Sampling Time 4 in Table 1. The represents the total number of marked individuals now in the population. Return all the beans from this second collection (which are now all marked, and therefor dark-colored) to the bag. Shake thoroughly.
9. Without looking, grab a handful of beans from the bag and place them on your desk. This represents your <u>fourth</u> capture of a group of organisms. Count the total number of beans you grabbed in this handful (regardless of whether or not they are marked) and record this number as the value for C_4 in Table 1.
   a. Count the number of beans that are marked. Record this number as your value for R_4 in Table 1.
10. Use the equation to calculate for N in Table 1.
11. Count the actual total number of beans in your bag. Record this number in Table 1.
12. Separate the beans back into their respective beakers.

Results: *Copy the data table into your lab notebook.*

Data Table 1

Sampling Time	# of Marked Individuals in the Population (M_i)	# of Individuals Captured (C_i)	# of Individuals Recaptured (R_i)
1			
2			
3			
4			
Estimated Population Size			
Calculated Population Size (N)			
Actual Population Size			

Post-Lab Discussion Questions

1. What would R equal if no marked individuals are recaptured? What might a researcher do if this happens?
2. If a marking technique was performed inadequately, meaning the mark washed off or fell off of animals once released back into the wild, how would that affect the researcher's calculated population size?
3. Would this method work for a population with a short life-cycle? Why or why not?
4. How would immigration/emigration affect the reliability of the calculated population size? Why?
5. Why was it important to not look in the bag as you were capturing your organisms (beans)?
6. Why was it important to shake the bag after releasing marked organisms back into the population?
7. How did your estimated population size compare to the actual population size?
8. How did the calculated population size (N) compare to the actual population size?
9. Overall, is the mark-recapture method a reliable way to estimate population size?
10. In general, what are 3 possible sources of error for the mark-recapture technique?